

INTRODUCTION TO DATABASES

Database Foundations

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2025-III



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FRANCISCO JOSÉ DE CALDAS

Outline

- 1 Software Components and Applications
- 2 DataBase Classification
- 3 Relational Database Design
- 4 DataBase Management Systems — DBMS
- 5 Data Engineering



Outline

MongoDB

1 Software Components and Applications

2 "Data Classification"

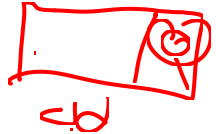
3 "Relational Database Design"

4 "Data Base Management Systems — DBMS"

5 Data Engineering

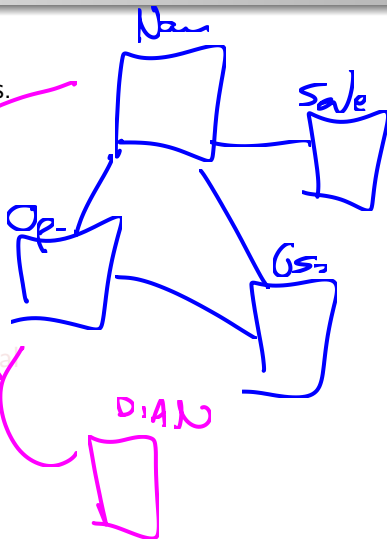
#cod #puede
#alfinorevise ~

"ud"



Modular Software Components

- **Software Components** are the building **blocks** of software systems.
- **Modular Software** is a software design technique that emphasizes **separating the functionality** of a program into independent, interchangeable **modules**.



<html>

~

<?php log_g \>

<p>~</p>



Modular Software Components

- **Software Components** are the building **blocks** of software systems.
- **Modular Software** is a software design technique that emphasizes **separating** the **functionality** of a program into independent, interchangeable **modules**.
- **Software Applications** are the **final product** of **software development**.
- **Software Development** is the process of **creating** software applications.

SDLC

→ req

→ ana

→ des

→ code

→ test

→ deploy

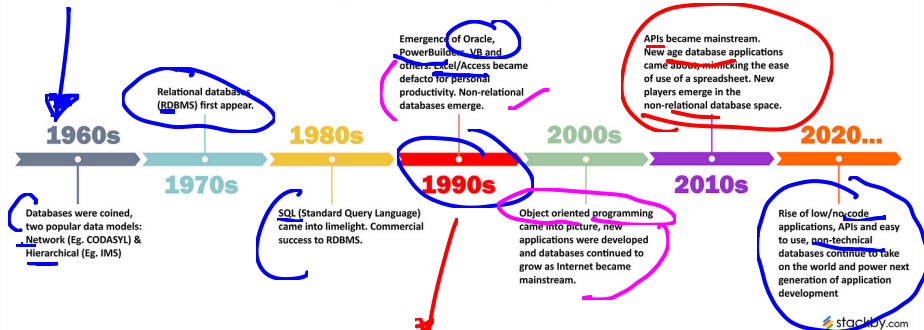
→ maintain

→ use

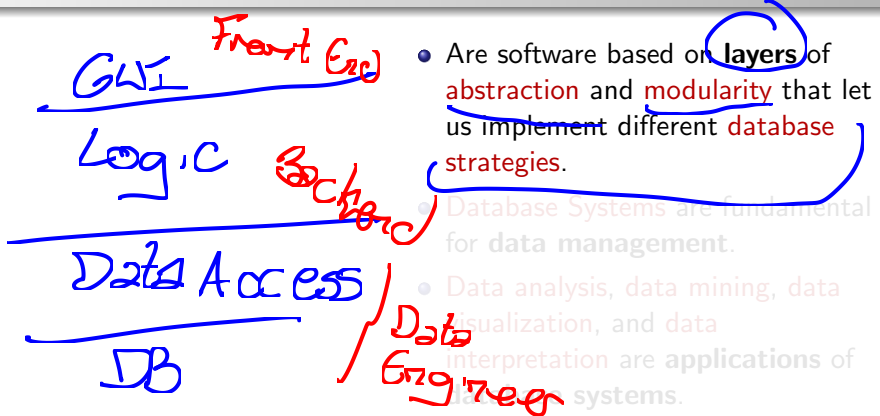


History of DataBases

History of Databases (1960-2020)



Applications



Applications

Store-Retrieve

DB

DB

DB

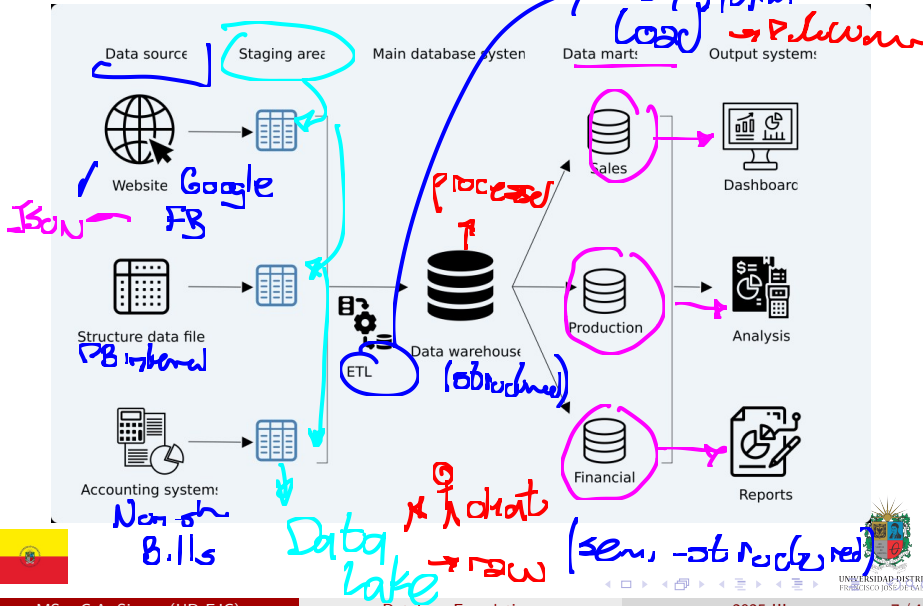
DB

- Are software based on **layers** of **abstraction** and **modularity** that let us implement different **database strategies**.
- **Database Systems** are fundamental for **data management**.
- **Data analysis, data mining, data visualization, and data interpretation** are **applications** of **database systems**.



Case Study: Database System

Extract → Data Lake
Transform → Data Warehouse
Load → Data Mart



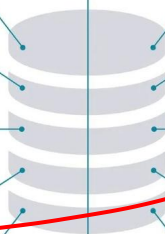
Structured and Unstructured Data

Structured Data

vs

Unstructured Data

Can be displayed
in rows, columns and
relational databases



| | | |
|----|----|----|
| XY | 1 | 2 |
| A | A1 | A2 |
| B | B1 | B2 |
| C | C1 | C2 |
| D | D1 | D2 |

Numbers, dates
and strings

0,1,2,3
3,4,5,6
6,7,8,9
DAY
MONTH
5, 2025 VZ,
month, 1
F=0-9

Estimated 20% of
enterprise data (Gartner)

20%

Requires less storage

••
••
••

Easier to manage
and protect with
legacy solutions



Cannot be displayed
in rows, columns and
relational databases



Images, audio, video,
word processing files,
e-mails, spreadsheets



Estimated 80% of
enterprise data (Gartner)

80%

Requires more storage

••
••
••
••
••
••
••
••

More difficult to
manage and protect
with legacy solutions



Tabular Structures

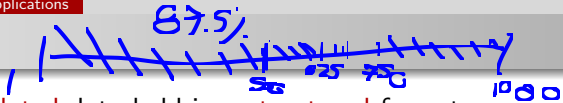
- **Table** is a collection of **related** data held in a **structured** format within a **database**.
- **Column** is a set of **data values** of a particular **simple type**, one for each row of the table.
- **Row** is a set of **data values** of a particular **relationship**, one for each column of the table.

- **Primary Key** is a unique identifier for a record in a data set.
- **Foreign Key** is a column or group of columns in a table that links to a primary key in another table.

| age | name | weight |
|-----|------|--------|
| 25 | John | 169 |



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- redundancy

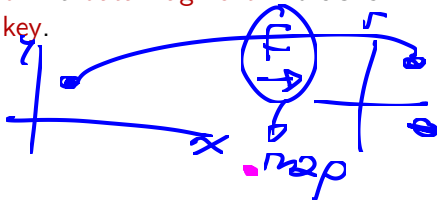
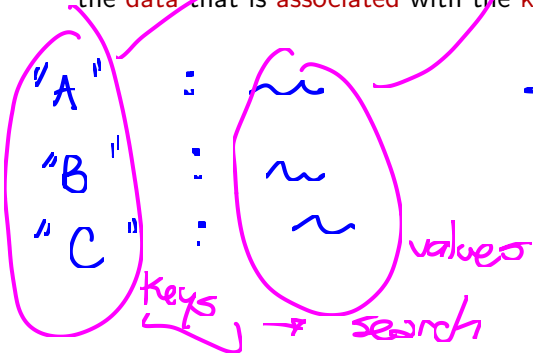
+ $\Rightarrow$ read $\Rightarrow$ index - PK



Key-Value Data Structures

Python - Dictionary
 JS - Object

- **Key-Value Data Structures** are a type of **data structure** that can map **keys** to **values**. $A = B$
- **Key** is a **unique** identifier for a **record** in a **data fragment**. **Value** is the **data** that is **associated** with the **key**.



CRUD Operations

- **CRUD** is an acronym for **Create**, **Read**, **Update**, and **Delete**.
- **Create** is the process of **adding** new **records** to a **data set**.
- **Read** is the process of **retrieving** records from a **data set**.
- **Update** is the process of **modifying** records in a **data set**.
- **Delete** is the process of **removing** records from a **data set**.

one of these

expense



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DataBase Classification

- **DataBase** is a collection of **data** that is **organized** so that it can be **easily accessed, managed, and updated**.
- Relational DataBase is a type of database that stores and provides access to data points that are related to one another.
- NoSQL DataBase is a type of database that provides a mechanism for storage and retrieval of data that is modeled in means other than the tabular relations used in relational databases.

Recall



DataBase Classification

- **DataBase** is a collection of **data** that is **organized** so that it can be **easily accessed, managed, and updated**.
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Analytics

Reports
Time Series



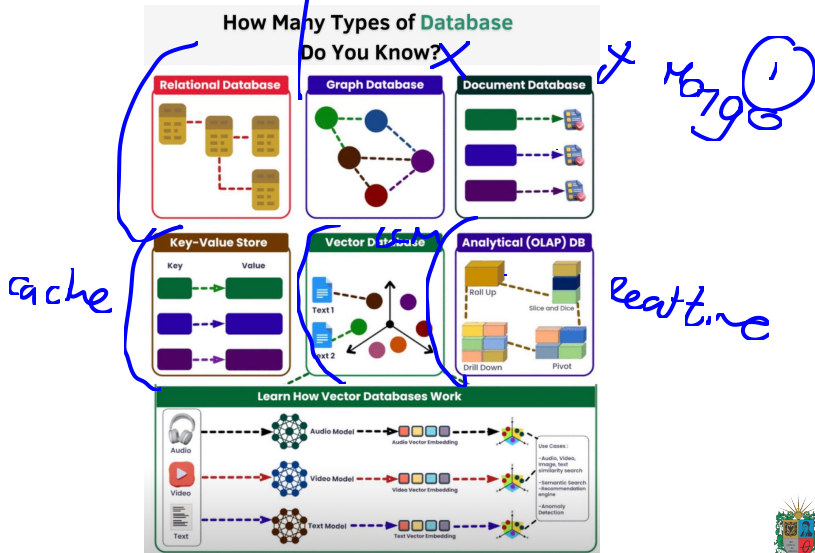
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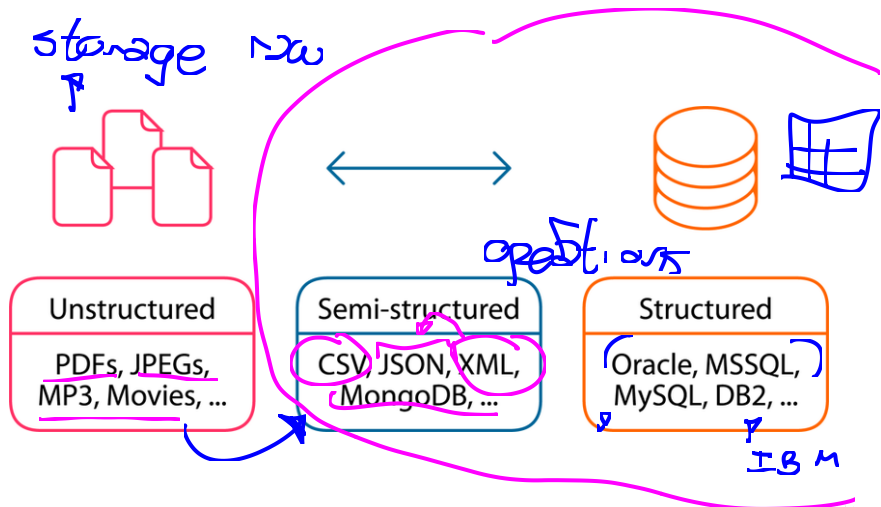
MongoDB — Functional
SQL



Types of Database



Semi-Structured Data



Relational Model

- The **relational model** is the **most common** and widely used model today.
 - It is based on the concept of **relations**. A **relation** is a table with **rows** and **columns**.
 - The **relational model** is based on the concept of **keys**, which leads to *strong relationships* in structured data.
 - It also incorporates the concepts of **integrity constraints** and **normalization**.
- client-side



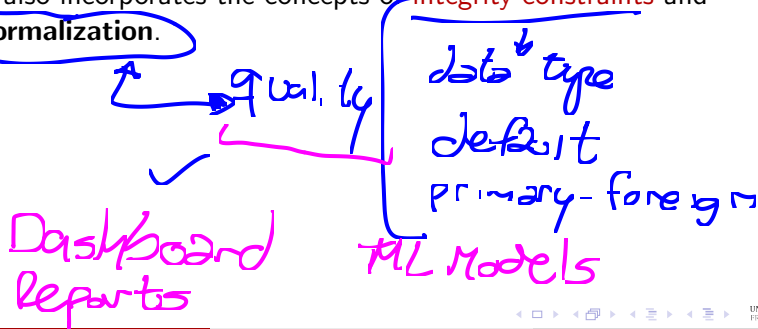
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Hierarchical Model

- The **hierarchical model** organizes data in a **tree-like structure**.
- It is based on the concept of **parent-child relationships**, meaning **one-to-many** relationships.
- An example of a **hierarchical model** is the **XML format**.

<list>

<student>

<name> Pepita </name>

<code> 23 </code>

</student>

<student>

<name> Pepito </name>

</student>

</list>

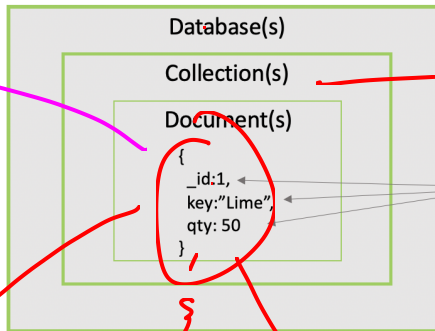


Document-Based Model

list of JSon



Instance
(primary)



MongoDB

list

Field(s)

schema
• mandatory
keys

-id

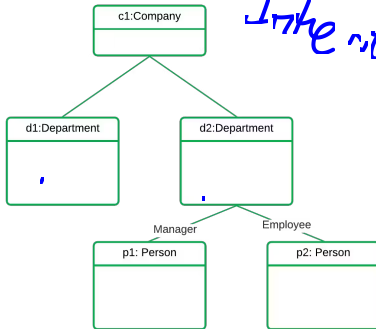
{
 _id:
 key:
 ,
 _id:
 qty:
}

key: value (tree)

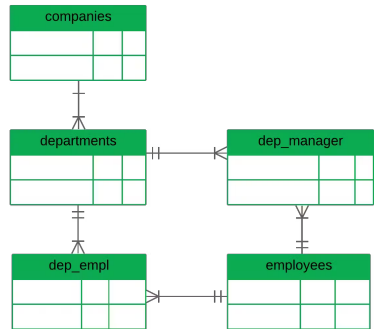


Object-Oriented Model

Object-Oriented



Relational

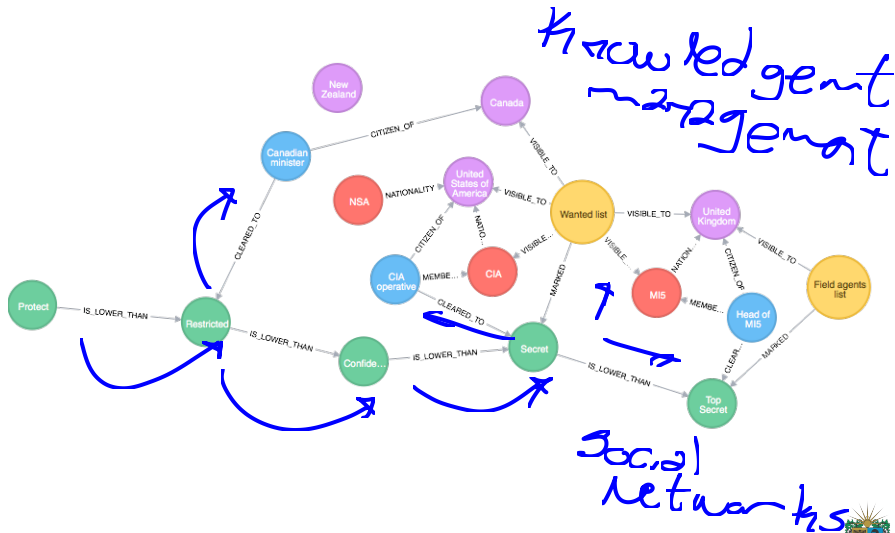


Instance

methods → store procedures



Graph-Based Model



NoSQL Model

What is NoSQL Database

High Scalability

High Availability

NoSQL Database

Big Data Capability

Fast Performance

Easy Replication

educba.com

V's
- Volume
- Variety
- Velocity



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Set Theory in Databases

- The **set theory** is a branch of **mathematical logic** that studies sets, which are **collections of objects**. row
- The **set theory** is applied in **databases** to define the **relational model** and the **relational algebra**.
- The **relational model** is a **mathematical model** of data for large **shared data banks** and it has a **solid theoretical foundation**.
- The **relational algebra** is a **procedural query language**, which takes **relations** as **input** and produces **relations** as **output**.



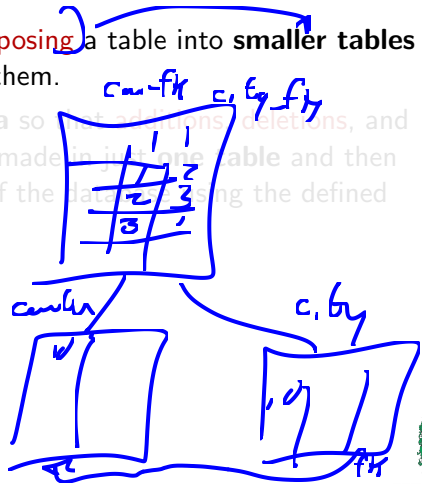
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Sentence $\rightarrow$ instruction



- Country City
- | | |
|---|---|
| 1 | 1 |
| 2 | 2 |
| 3 | 3 |



Normalization in Databases

- **Normalization** is the process of **organizing** the **columns** (attributes) and **tables** (relations) of a relational database to **minimize data redundancy**.
- **Normalization** involves **decomposing** a table into **smaller tables** and defining **relationships** between them.
- The objective is to **isolate data** so that **additions, deletions, and modifications** of a field can be made in just **one table** and then **propagated** through the rest of the database using the defined relationships.

↙
Cascade



Definitions

- Entity**: A thing or **object** in the real world that is **distinguishable** from other **objects**.

logic
 ↳
 list of objects
- Attribute**: A property or characteristic of an entity.
- Relationship**: An association between entities.
- Cardinality**: The number of instances of an entity that can be associated with another entity.

list of objects

list of rows
- Degree**: The number of entities that participate in a relationship.



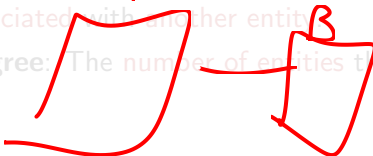
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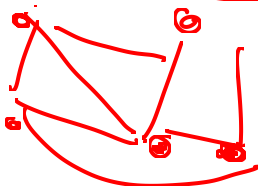
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Entity-Relation Model

- The **Entity-Relation Model** is a graphical representation of the entities and their relationships in a database.
- The **Entity-Relation Model** is used to design the schema of a database and to communicate the design to stakeholders.

students

name Text (255)
code int primary



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What is a DBMS?

- A **Database Management System** (DBMS) is a **software system** that uses a **standard** method to **store** and **organize data**.
- A **DBMS** is a **software system** that allows users to **define**, **create**, **maintain**, and **control access** to the database.
- A **DBMS** is a **software package** designed to **manipulate**, **retrieve**, and **manage data** in a database.



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Pros & Cons of DBMS

● Pros:

- **Data Independence:** Data is **stored independently** of the applications that use it.
- **Data Integrity:** Data is **consistent** and **accurate**.
- **Data Security:** Data is **protected** from **unauthorized access**.
- **Data Recovery:** Data can be **recovered** in case of **failure**.

● Cons:

- **Complexity:** DBMS are complex systems.
- **Cost:** Cost: DBMS can be expensive for large data volumes.
- **Performance:** DBMS can be slow.



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GUI Assistants

- A **Graphical User Interface** (GUI) is a type of user interface that allows **users** to **interact** with electronic devices using **graphical icons** and **visual indicators**.
- GUIs are easier to use than Command Line Interfaces (CLI) because they allow **users** to **interact** with the system using **visual elements** such as **windows**, **buttons**, and **menus**.
- GUIs are more **intuitive** and **user-friendly** than CLIs, which makes them ideal for **users** who are **not familiar** with the system.



Case Study: DBeaver

The screenshot shows the DBeaver 22.1.3 interface. On the left, the 'Database Navigator' pane shows a tree view of the database structure: system - localhost:26257 > Databases > movr > public > Tables > rides. The main panel displays the 'Properties' tab for the 'rides' table. The 'Table Name' is 'rides', 'Object ID' is '111', and 'Owner' is 'root'. Below this, the 'Columns' tab shows a list of columns with their data types and constraints.

| Column Name | # | Data type | Identity | Collation | Not Null | Default | Comm |
|---------------|----|----------------|----------|-----------|----------|---------|------|
| id | 1 | uuid | | | [v] | | |
| city | 2 | varchar | | default | [v] | | |
| vehicle_city | 3 | varchar | | default | [] | | |
| rider_id | 4 | uuid | | | [] | | |
| vehicle_id | 5 | uuid | | | [] | | |
| start_address | 6 | varchar | | default | [] | | |
| end_address | 7 | varchar | | default | [] | | |
| start_time | 8 | timestamp | | | [] | | |
| end_time | 9 | timestamp | | | [] | | |
| revenue | 10 | numeric(10, 2) | | | [] | | |

At the bottom, the 'Project - General' pane shows the 'DataSource' tab with 'Name' and 'DataSource' fields. The status bar at the very bottom indicates 'PST en_US'.



Command Line

- A **Command Line Interface** (CLI) is a type of **user interface** that allows **users** to **interact** with electronic devices using **text-based commands**.
- CLIs are more **powerful** and **flexible** than GUIs because they allow users to perform **complex tasks** using **simple commands**.
- CLIs are more **efficient** than GUIs because they do **not require** users to **navigate** through menus and windows to perform tasks.



Case Study: MariaDB CLI

```

arnel@arnel.com [~]# mysql -u arnel_test2 -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 8643
Server version: 10.1.25-MariaDB MariaDB Server

Copyright (c) 2000, 2017, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| arnel_test1 |
| arnel_test2 |
| information_schema |
+-----+
3 rows in set (0.00 sec)

MariaDB [(none)]> use arnel_test1
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [arnel_test1]>
    
```



Why use an agnostic tool?

- An **agnostic tool** is a tool that is **not tied** to a specific **technology** or **platform**.
- **Agnostic tools** are useful because they **allow users** to work with **multiple databases** without having to **learn different tools**.
- **Agnostic tools** are also useful because they **allow users** to work with **multiple databases** without having to **switch between different tools**.



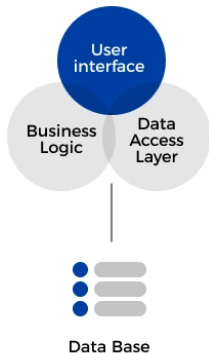
Microservices Architecture

- **Microservices Architecture** is a **software architecture** in which the **components** of the software are **broken down** into **small, independent services**.
- **Microservices Architecture** is a **modern software architecture** that is used to **build large** and **complex software systems**.
- **Microservices Architecture** is a **flexible** and **scalable software architecture** that is used to **build software systems** that require high scalability and flexibility.

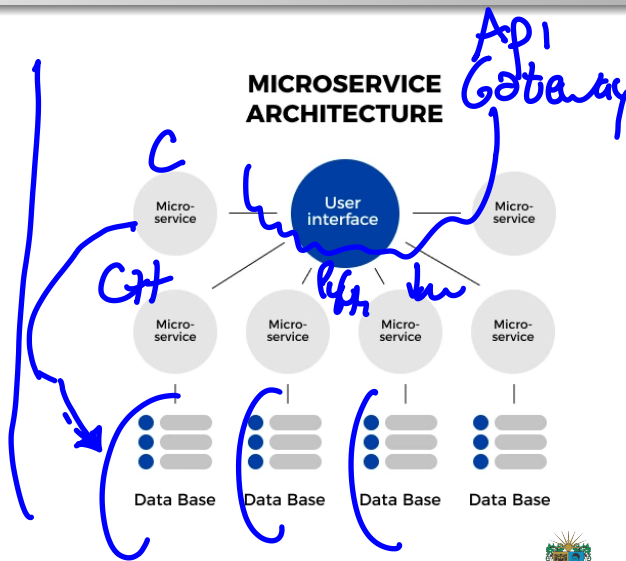


Monolithic Architecture Schema

MONOLITHIC ARCHITECTURE



MICROSERVICE ARCHITECTURE



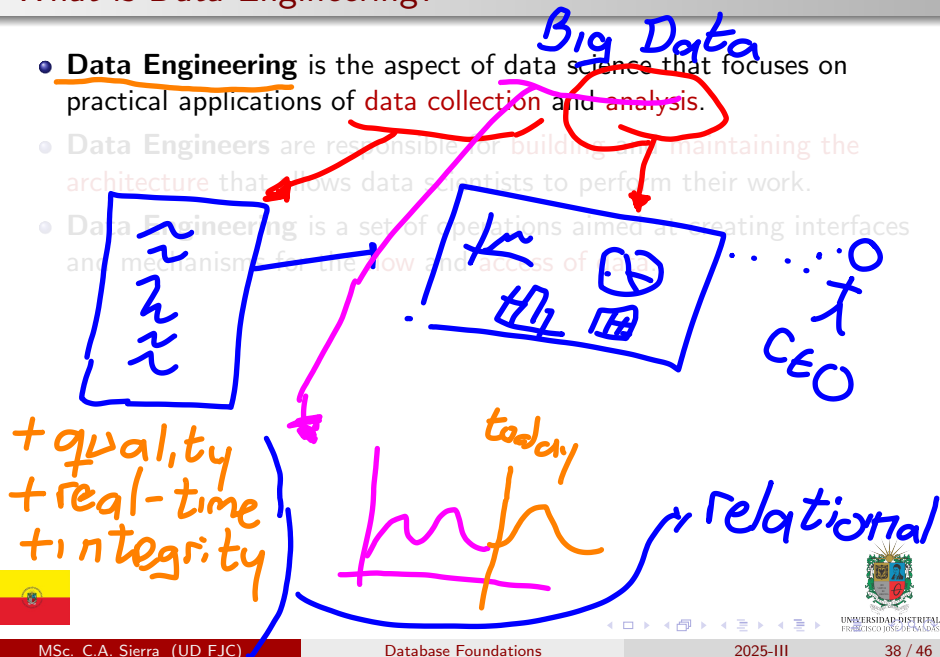
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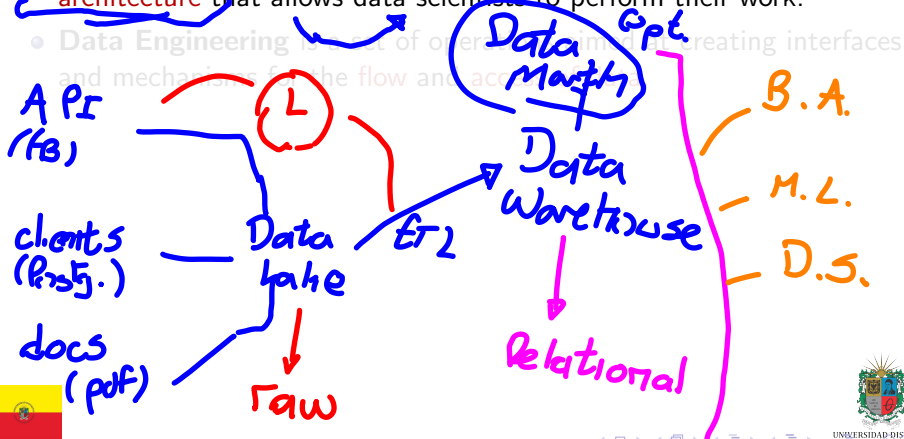
What is Data Engineering?

- Data Engineering is the aspect of data science that focuses on practical applications of data collection and analysis.
- Data Engineers are responsible for building and maintaining the architecture that allows data scientists to perform their work.
- Data Engineering is a set of operations aimed at creating interfaces and mechanisms for the flow and access of data.



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• observability → DevOps



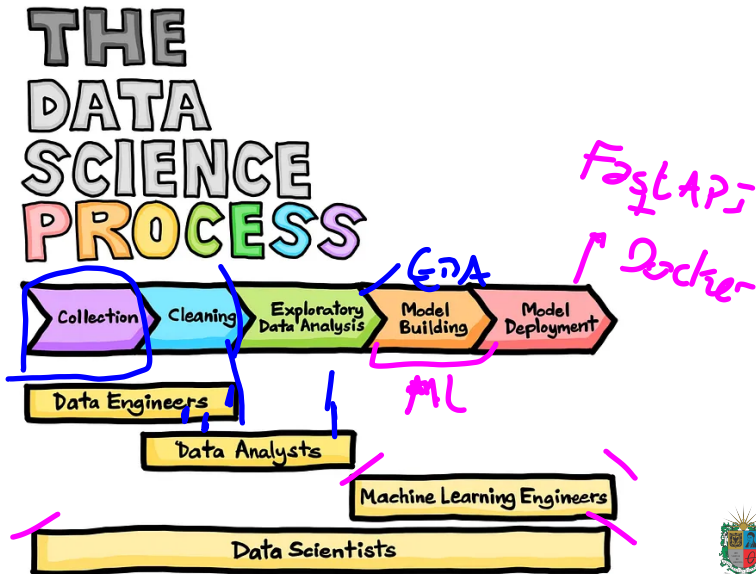
Why is Data Engineering Important?

- **Data Engineering** is the foundation of the **high-quality data** that is necessary for **effective data science**.
- **Data Engineering** is the process of **collecting**, **transforming**, and **storing data** in a way that's accessible and easy to analyze.

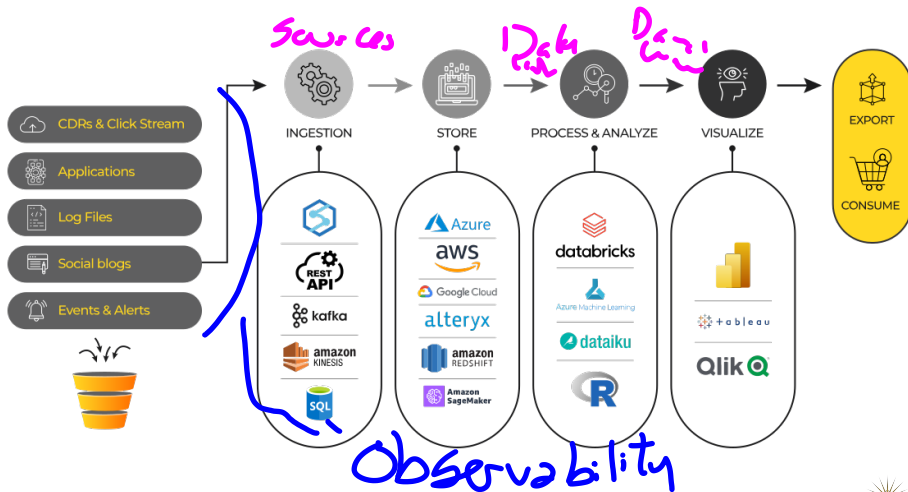
sources → ETL → Persistence



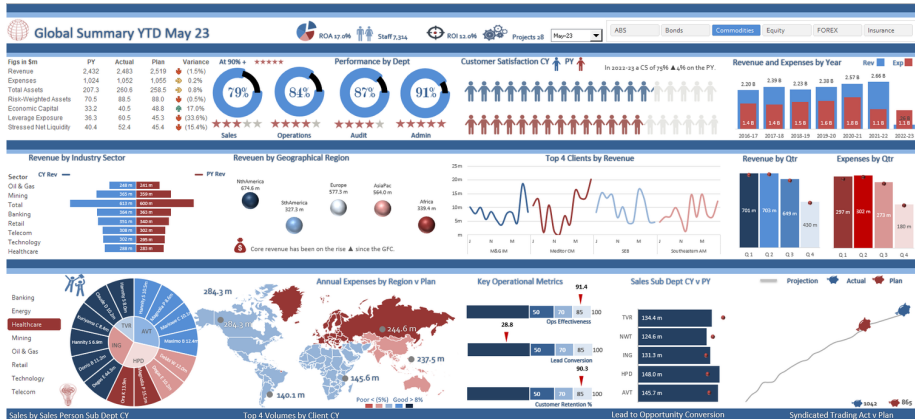
Data Science Workflow



Data Engineering Architecture



Case Study: Dashboards



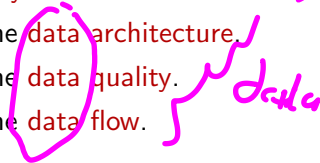
DBOps vs Data Engineer

- **DBOps** is responsible for the operation of the database.
- **DBOps** is responsible for the performance of the database.
- **DBOps** is responsible for the security of the database.
- Data Engineer is responsible for the data architecture.
- Data Engineer is responsible for the data quality.
- Data Engineer is responsible for the data flow.



DBOps vs Data Engineer

- **DBOps** is responsible for the **operation** of the database.
- **DBOps** is responsible for the **performance** of the database.
- **DBOps** is responsible for the **security** of the database.
- **Data Engineer** is responsible for the **data** architecture.
- **Data Engineer** is responsible for the **data** quality.
- **Data Engineer** is responsible for the **data** flow.



How to improve data quality?

- **Data Quality** is the process of ensuring that data is accurate, complete, and reliable.
- **Data Quality** is the process of ensuring that data is consistent and up-to-date.
- **Data Quality** is the process of ensuring that data is free from errors and inconsistencies.
- **Data Quality** is the process of ensuring that data is of high quality and can be trusted.
- **Data Quality** is the process of ensuring that data is fit for purpose and can be used effectively.



Outline

- 1 Software Components and Applications
- 2 DataBase Classification
- 3 Relational Database Design
- 4 DataBase Management Systems — DBMS
- 5 Data Engineering



Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/databases-ii>

